



Eco-1: A Z-JEPA Hyper-Modal MoE Architecture with Active Infer- ence Alignment

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Abstract

We present **Eco-1**, a 72B-parameter hyper-modal architecture whose core is **Z-JEPA** (Zoo Joint-Embedding Predictive Architecture): a cross-modal prediction system that learns world-state dynamics in latent space across video, audio, text, and 3D point clouds. Eco-1 composes three subsystems: (1) a Z-JEPA backbone that extends Meta’s V-JEPA from single-modality video to arbitrary modality combinations via shared latent prediction targets; (2) an HLLM (Hamiltonian Large Language Model) that governs sparsely-activated MoE routing through energy-conserving dynamics with Hidden Markov gating; and (3) a planning prior derived from Expected Free Energy (EFE) minimization, which unifies goal pursuit with epistemic exploration under Friston’s active inference framework. The MoE layer routes through 8 behavioral experts (hunting, grazing, flocking, nesting, territorial, mating, migration, playing) trained on 10M hours of virtualized animal ecosystem video. We prove that EFE-derived gating produces load-balanced expert utilization with bounded variance, and that the Hamiltonian backbone ensures inference stability across 128K-token multimodal contexts. Eco-1 achieves 89.4% on Perception Test, 91.2% behavioral prediction accuracy, and 0.847 cross-modal retrieval mAP, while serving as the foundation for Zoo’s avatar tutors, conservation agents, and personalized AI.

Keywords: joint-embedding predictive architecture, mixture of experts, active inference, multimodal learning, Hamiltonian dynamics

1 Introduction

Current multimodal AI systems process video, audio, and text through modality-specific encoders connected by learned projection layers. This design has two structural limitations. First, the connection between modalities is learned post-hoc rather than through a shared predictive objective, leading to brittle cross-modal transfer. Second, the models are passive: they respond to inputs but do not plan actions to reduce their own uncertainty about the world.

Eco-1 addresses both limitations. Z-JEPA defines a shared latent space where *any* modality can predict *any other* modality’s representation, trained via self-supervised masking without pixel or waveform reconstruction. The active inference component introduces a planning prior based on Expected Free Energy (EFE) [5], which naturally balances exploitation (achieving goals) with exploration (reducing uncertainty).

1.1 Architecture Overview

Eco-1 is a composition of three subsystems, each with a distinct role:

1. **Z-JEPA** (Section 2): Cross-modal latent prediction. Learns shared representations by predicting masked modality patches from visible ones.
2. **HLLM backbone** (Section 3): Hamiltonian reasoning engine with MoE routing. Provides energy-conserving inference dynamics and stable expert selection.
3. **EFE planning** (Section 4): Active inference alignment. Selects actions (including which experts to activate and what to attend to) by minimizing expected free energy.

2 Z-JEPA: Cross-Modal Prediction in Latent Space

2.1 Motivation: Why Not Pixels?

Generative approaches to multimodal learning (e.g., predicting future video frames in pixel space) waste capacity modeling high-frequency noise. V-JEPA [1] demonstrated that self-supervised video

Modality	Encoder	Patch Size	Dim
Video	ViT-Huge/14	(2, 16, 16)	1280
Audio	Audio Spectrogram Transformer	(2, 128)	768
Text	Learned BPE embeddings	1 token	4096
3D	Point Transformer v3	1024 points	512

Table 1: Modality-specific encoders in Z-JEPA.

understanding in *latent* space—predicting the representation of masked patches rather than the pixels—produces stronger downstream performance with less compute.

Z-JEPA extends this principle across modalities. The core insight: if modality A and modality B share underlying world-state dynamics, then predicting B ’s latent representation from A ’s latent representation is both more efficient and more robust than predicting B ’s raw signal from A ’s raw signal.

2.2 Formal Definition

Definition 2.1 (Z-JEPA Objective). *Let $\mathcal{M} = \{m_1, \dots, m_K\}$ be a set of modalities. For each modality m_k , let $f_k : \mathcal{X}_k \rightarrow \mathbb{R}^d$ be the encoder and $\bar{f}_k$ its exponential moving average (target encoder). The Z-JEPA objective is:*

$$\mathcal{L}_{Z-JEPA} = \sum_{i \neq j} \mathbb{E}_{x_i, x_j} [\|g_\psi(f_i(x_i^{vis})) - \bar{f}_j(x_j^{mask})\|^2] \quad (1)$$

where x_i^{vis} denotes the visible patches of modality i , x_j^{mask} denotes the masked patches of modality j , and g_ψ is a cross-modal predictor.

The target encoders $\bar{f}_k$ are updated via exponential moving average with momentum $\tau = 0.998$:

$$\bar{f}_k \leftarrow \tau \bar{f}_k + (1 - \tau) f_k$$

This prevents representation collapse without requiring contrastive negatives [14].

2.3 Modality Encoders

All encoders project to a shared latent dimension $d = 2048$ via learned linear projections before the cross-modal predictor g_ψ .

2.4 Masking Strategy

For video, we follow V-JEPA [1] with 90% masking ratio over spatiotemporal tubes. For audio, 80% of spectrogram patches are masked. For text, 15% of tokens (following BERT-style masking [15]). For 3D, 75% of point groups. The aggressive masking forces the model to learn high-level structure rather than local correlations.

Theorem 2.1 (Cross-Modal Prediction Consistency). *Under the Z-JEPA objective with shared latent dimension d and target encoder momentum $\tau \rightarrow 1$, the learned representations $f_i(x_i)$ and $f_j(x_j)$ for co-occurring signals x_i, x_j satisfy:*

$$\|f_i(x_i) - f_j(x_j)\|^2 \leq \frac{2\mathcal{L}_{Z-JEPA}}{1 - \tau} + O((1 - \tau)^2)$$

That is, cross-modal representations converge as the training loss decreases.

Expert	Behavioral Class	Transfer Domain
E_1 : Hunting	Predator strategies	Planning, optimization
E_2 : Grazing	Resource harvesting	Steady-state operations
E_3 : Flocking	Group coordination	Multi-agent systems
E_4 : Nesting	Construction/creation	Generative tasks
E_5 : Territorial	Space management	Resource allocation
E_6 : Mating	Social bonding	Relationship modeling
E_7 : Migration	Long-range planning	Sequential decision making
E_8 : Playing	Exploration/learning	Active inference

Table 2: Behavioral experts and their transfer domains.

3 HLLM Backbone with MoE Routing

The language modeling and reasoning component uses the HLLM architecture [12]. We summarize the key elements and describe the MoE integration.

3.1 MoE Configuration

Eco-1 uses 8 behavioral experts, each specialized for a class of dynamic patterns observed in animal ecosystems:

Each expert is a feed-forward network of the form:

$$E_k(x) = W_2^{(k)} \cdot \text{SwiGLU}(W_1^{(k)}x) \quad (2)$$

with $W_1^{(k)} \in \mathbb{R}^{d_{\text{ff}} \times d}$, $W_2^{(k)} \in \mathbb{R}^{d \times d_{\text{ff}}}$, and $d_{\text{ff}} = 28672$.

3.2 Hamiltonian Gating

The gating mechanism uses the momentum vector p_t from the HLLM leapfrog dynamics (see [12] for full derivation). Each expert k has a learned momentum centroid $\mu_k \in \mathbb{R}^d$. The routing probability is:

$$\pi(z_t = k \mid z_{t-1} = j, p_t) = \frac{\exp(-\beta \|p_t - \mu_k\|^2) \cdot T_{jk}}{\sum_{k'} \exp(-\beta \|p_t - \mu_{k'}\|^2) \cdot T_{jk'}} \quad (3)$$

where $T \in \mathbb{R}_{\geq 0}^{K \times K}$ is the HMM transition matrix and $\beta > 0$ is the inverse temperature.

4 Active Inference Alignment via EFE

4.1 Free Energy Principle

Following Friston [6], we model Eco-1 as an agent that maintains a generative model of its environment and acts to minimize the divergence between its predictions and observations. The variational free energy is:

$$F = \text{KL}[Q(s) \parallel P(s)] - \mathbb{E}_{Q(s)}[\log P(o \mid s)] \quad (4)$$

where $Q(s)$ is the approximate posterior (beliefs), $P(s)$ is the prior, and $P(o \mid s)$ is the likelihood.

4.2 Expected Free Energy for Action Selection

The Expected Free Energy (EFE) for a policy π evaluates future actions:

$$G(\pi) = \underbrace{-\mathbb{E}_{Q(\tilde{o}|\pi)}[\log P(\tilde{o})]}_{\text{pragmatic value}} + \underbrace{\mathbb{E}_{Q(\tilde{o}|\pi)}[H[Q(\tilde{s} \mid \tilde{o}, \pi)]]}_{\text{epistemic value}} \quad (5)$$

Definition 4.1 (EFE-Aligned Expert Selection). *Given the current belief state $Q(s)$ and candidate expert-policy pairs $\{(\pi_k, E_k)\}_{k=1}^K$, the EFE-aligned selection is:*

$$k^* = \arg \min_k G(\pi_k) \quad \text{s.t. load balance constraints}$$

Theorem 4.1 (EFE Load Balance). *Let $\alpha \in (0, 1)$ be the epistemic weight in EFE computation. Under EFE-aligned expert selection with N tokens per batch and K experts:*

$$\mathbb{E} \left[\left(\bar{g}_k - \frac{1}{K} \right)^2 \right] \leq \frac{1 - \alpha}{K^2} + \frac{1}{NK}$$

where $\bar{g}_k = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[k_i^* = k]$ is the empirical load on expert k .

Proof. The epistemic term in EFE acts as an entropy bonus over expert selection: maximizing information gain is equivalent to maximizing the entropy of the belief update, which distributes probability mass across experts that reduce uncertainty in different dimensions. With weight α on the epistemic term, the effective gating distribution has entropy at least $\alpha \log K$. The variance bound follows from the maximum-entropy characterization: a distribution with entropy $\alpha \log K$ has per-category variance at most $(1 - \alpha)/K^2$. The $1/(NK)$ term is the finite-sample correction. $\square$

4.3 Latent Rollouts for Planning

Eco-1 performs planning in Z-JEPA’s latent space rather than in observation space. Given current latent state z_t and candidate action a , the predicted next state is:

$$\hat{z}_{t+1} = g_\psi(z_t, a) \quad (6)$$

EFE is computed over these latent rollouts:

5 Training Pipeline

5.1 Data

5.2 Four-Stage Training

1. **Stage 1: V-JEPA pretraining** (30 days, 256 H100s). Self-supervised video prediction following V-JEPA 2 [2]. Objective: Equation 1 restricted to video-to-video prediction.
2. **Stage 2: Language modeling** (45 days, 512 H100s). Next-token prediction with thinking tokens. HLLM dynamics with energy conservation loss.
3. **Stage 3: Cross-modal alignment** (15 days, 128 H100s). Full Z-JEPA objective over all modality pairs (Equation 1).
4. **Stage 4: Behavioral specialization** (10 days, 64 H100s). MoE expert routing optimization via group sequence optimization [12].

Algorithm 1 EFE Latent Planning in Eco-1

```
1: Input: Current latent state  $z_t$ , action set  $\mathcal{A}$ , horizon  $H$ 
2: Output: Action with minimum EFE
3: for each action sequence  $\pi = (a_1, \dots, a_H) \in \mathcal{A}^H$  do
4:    $z \leftarrow z_t$ 
5:    $G_\pi \leftarrow 0$ 
6:   for  $h = 1$  to  $H$  do
7:      $\hat{z} \leftarrow g_\psi(z, a_h)$  {Latent prediction}
8:      $G_\pi \leftarrow G_\pi - \log P(\hat{z})$  {Pragmatic}
9:      $G_\pi \leftarrow G_\pi + H[Q(s \mid \hat{z})]$  {Epistemic}
10:     $z \leftarrow \hat{z}$ 
11:   end for
12: end for
13: return  $\pi^* = \arg \min_\pi G_\pi$ 
```

Data Source	Volume	Modalities
Virtualized ecosystems	10M hours video	Video, audio, 3D
Web text corpora	10T tokens	Text
Scientific papers	100B tokens	Text
Image-text pairs	1B pairs	Image, text
Video-text pairs	100M pairs	Video, text
Camera trap footage	2M hours	Video

Table 3: Training data for Eco-1.

5.3 BitDelta Personalization

After training, Eco-1 supports per-user personalization via BitDelta [13]. The personalized model is:

$$W_{\text{user}} = W_{\text{base}} + \gamma_u \cdot \text{sign}(\delta_u) \quad (7)$$

where δ_u is the user-specific fine-tuning delta and γ_u is a per-tensor scale. This requires 0.7GB per user versus 144GB for the full model.

6 Experiments

6.1 Multimodal Understanding

6.2 Behavioral Prediction

Eco-1’s behavioral experts are evaluated on predicting the next action in virtualized ecosystem sequences:

Expert utilization ranges from 11.3% to 14.2% across 8 experts (ideal: 12.5%), confirming the load balance guarantee of Theorem 4.1.

Benchmark	Eco-1	GPT-4V	Gemini 1.5
Perception Test (video QA)	89.4%	84.1%	86.7%
Cross-modal retrieval (mAP)	0.847	0.791	0.823
Video captioning (CIDEr)	142.3	131.7	138.9
3D scene understanding	87.2%	–	–

Table 4: Multimodal understanding benchmarks.

Behavior Class	Prediction Accuracy	Expert Utilization
Hunting	93.7%	14.2%
Grazing	95.1%	11.8%
Flocking	91.8%	13.6%
Nesting	89.3%	12.1%
Territorial	90.6%	12.9%
Mating	88.4%	11.3%
Migration	87.9%	12.7%
Playing	92.8%	11.4%
Overall	91.2%	balanced

Table 5: Behavioral prediction accuracy and expert load balance.

6.3 Active Inference Planning

We evaluate the EFE planning component on navigation tasks in 3D environments:

The EFE planner with $\alpha = 0.5$ (equal epistemic/pragmatic weight) achieves the highest goal-reach rate, while $\alpha = 0.7$ (more exploration) achieves higher exploration efficiency at a small cost to goal completion.

7 Transfer to Downstream Applications

Eco-1’s behavioral experts transfer to application domains:

- **Avatar tutors** (ZIP-0008): The “playing” expert (E_8) provides the exploration-driven teaching style. The “migration” expert (E_7) handles long-range lesson planning.
- **Conservation agents** (ZIP-0011): The “hunting” expert (E_1) models poaching behavior for anti-poaching planning. The “flocking” expert (E_3) models herd dynamics for population monitoring.
- **Game NPCs**: The full expert roster maps directly to NPC behavioral archetypes.
- **DeFi advisors**: The “hunting” expert (E_1) transfers to arbitrage detection. The “territorial” expert (E_5) transfers to liquidity range management.

Planning Method	Goal Reach Rate	Exploration Efficiency
Random	23.1%	1.0×
Greedy (no epistemic)	67.4%	1.8×
EFE ($\alpha = 0.5$)	85.2%	3.2×
EFE ($\alpha = 0.7$)	81.9%	4.1×

Table 6: Active inference planning performance. Exploration efficiency is relative to random.

8 Related Work

Joint-embedding architectures. JEPA [3] was proposed by LeCun as a path to autonomous machine intelligence. V-JEPA [1] implements it for video. V-JEPA 2 [2] scales it further. I-JEPA [4] applies the principle to images. Z-JEPA extends to arbitrary modality combinations with a shared predictor.

Active inference in AI. Friston’s free energy principle [6] and its application to action selection via EFE [7] have been explored in robotics [8] but not in large-scale multimodal models. Eco-1 is, to our knowledge, the first system to integrate EFE planning at the scale of a 72B-parameter model.

MoE architectures. Mixtral [9], Switch Transformer [10], and GShard [11] demonstrate efficient MoE training. Eco-1 distinguishes itself through Hamiltonian gating and behavioral specialization derived from animal ecosystem data.

9 Conclusion

Eco-1 unifies three ideas—cross-modal latent prediction (Z-JEPA), energy-conserving inference (HLLM), and active inference planning (EFE)—into a single 72B-parameter architecture. The behavioral experts trained on virtualized ecosystems transfer to practical domains including education, conservation, gaming, and finance. The formal guarantees on load balance (Theorem 4.1), cross-modal consistency (Theorem 2.1), and inference stability (via HLLM’s energy conservation) provide the theoretical foundation for deploying Eco-1 as the backbone of Zoo’s decentralized AI infrastructure.

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